Project Plan

Three-Week Implementation Schedule

2.1 Project Timeline

The project will be completed within **three weeks**, following an accelerated development schedule. Tasks will be performed in parallel whenever possible to reduce implementation time and allow early integration and testing.

Phase	Timeline	Main Activities	Expected Deliverables
1. Requirements & Architecture	Days 1–2	Finalize requirements, define the MVP scope, design the system architecture, and agree on API contracts and database schemas	Approved requirements, architecture diagram, and database design
2. Core Development	Days 3–7	Implement person detection and single-camera tracking; develop the initial backend, database, and dashboard structure in parallel	Working CV pipeline, initial API, database, and dashboard skeleton
3. Feature Implementation	Days 8–11	Implement zone monitoring, event rules, incident management, evidence storage, and dashboard integration	Functional tracking and incident management features
4. System Integration	Days 12–14	Connect all components, test end-to-end workflows, and resolve integration issues	Integrated end-to-end MVP
5. Testing & Optimization	Days 15–17	Test detection, tracking, event accuracy, API performance, error handling, and camera disconnection scenarios	Test results, performance measurements, and bug fixes

6. AWS Deployment	Days 18–19	Deploy the application, configure access permissions, storage, logging, and basic cost monitoring	Working AWS deployment
7. Final Validation & Delivery	Days 20–21	Perform acceptance testing, complete documentation, prepare the demonstration, and finalize the presentation	Final application, technical documentation, testing report, and presentation

2.2 Key Milestones

- **Day 2:** Requirements and architecture finalized.
- **Day 7:** Core Computer Vision pipeline and initial application components implemented.
- **Day 11:** Main tracking, zone monitoring, and incident features completed.
- **Day 14:** End-to-end integration completed.
- **Day 17:** Testing and bug fixing completed.
- **Day 19:** AWS deployment completed.
- **Day 21:** Final acceptance, documentation, and project presentation completed.

2.3 Required Resources

Human Resources

- Project Manager / Team Coordinator.
- Computer Vision / Machine Learning Developer.
- Backend Developer.
- Frontend Developer.
- Cloud / DevOps and Testing Engineer.

Team members may cover multiple roles depending on the team size.

Technical Resources

- Python, OpenCV, a pretrained YOLO model, and ByteTrack.
- FastAPI and PostgreSQL.
- React or an equivalent frontend framework.
- AWS EC2, S3, IAM, and CloudWatch; Amazon RDS if the approved budget permits.
- Git and GitHub for version control.
- A local computer or Google Colab for initial development and testing.

Schedule Management

Because the project duration is limited to three weeks, the team will prioritize core MVP functionality and avoid adding nonessential features during implementation. Daily progress checks and early integration will help identify blockers before they affect the final delivery date.